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Class: T21

EXPERIMENT NO. 08

Aim: CI/CD Pipeline to host a website on AWS

Theory:

CI/CD (Continuous Integration and Continuous Deployment) is a software development practice that automates the process of integrating code changes, testing applications, and deploying them to a production environment. It helps developers deliver software faster, reduce human errors, and maintain application quality.

Continuous Integration (CI) is the process in which developers frequently push their code to a shared repository, such as GitHub. Whenever new code is pushed, an automated workflow is triggered to build the application and run tests to identify errors.

Continuous Deployment (CD) is the process of automatically deploying successfully tested code to a hosting environment. This ensures that the latest version of the website is available to users without requiring manual deployment for every change.

AWS Services Used

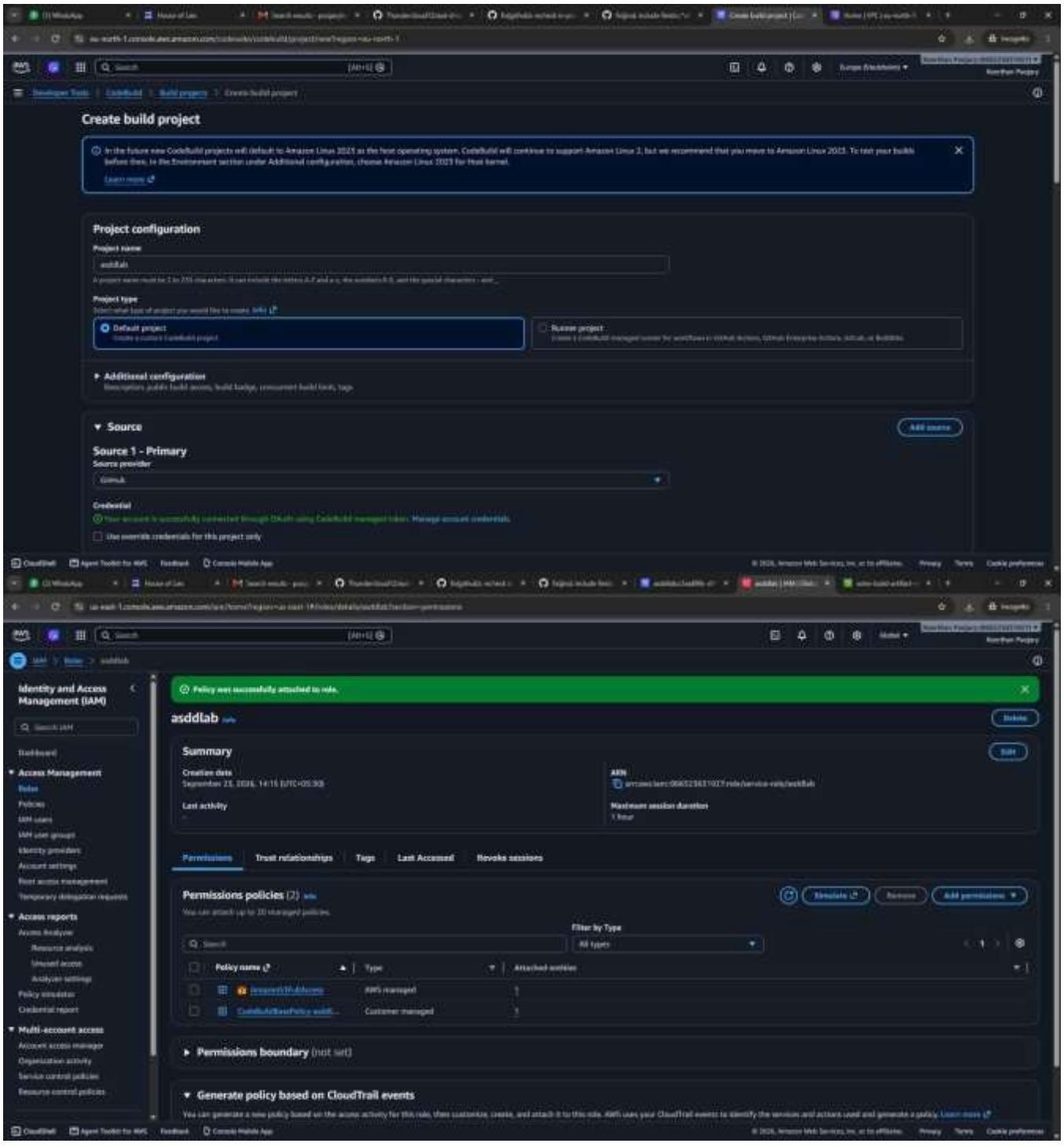
1. Amazon Web Services (AWS): A cloud computing platform that provides infrastructure and services for hosting websites and applications.
2. Amazon S3: A storage service that can host static websites containing HTML, CSS, JavaScript, and other static files.

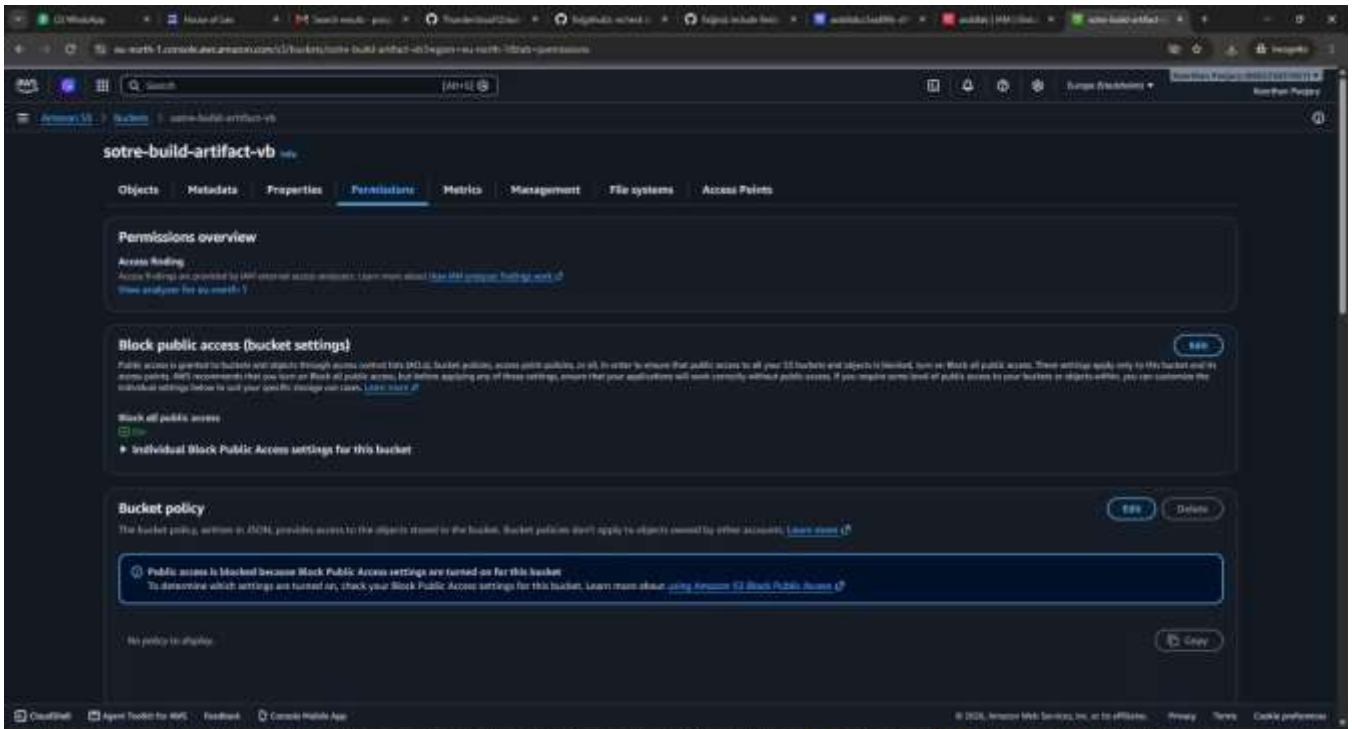
3. GitHub: A version control platform used to store source code and track changes.
4. GitHub Actions: An automation tool used to create CI/CD workflows that build, test, and deploy the website.
5. AWS IAM: Identity and Access Management controls access to AWS resources and allows the deployment workflow to receive only the permissions it needs.

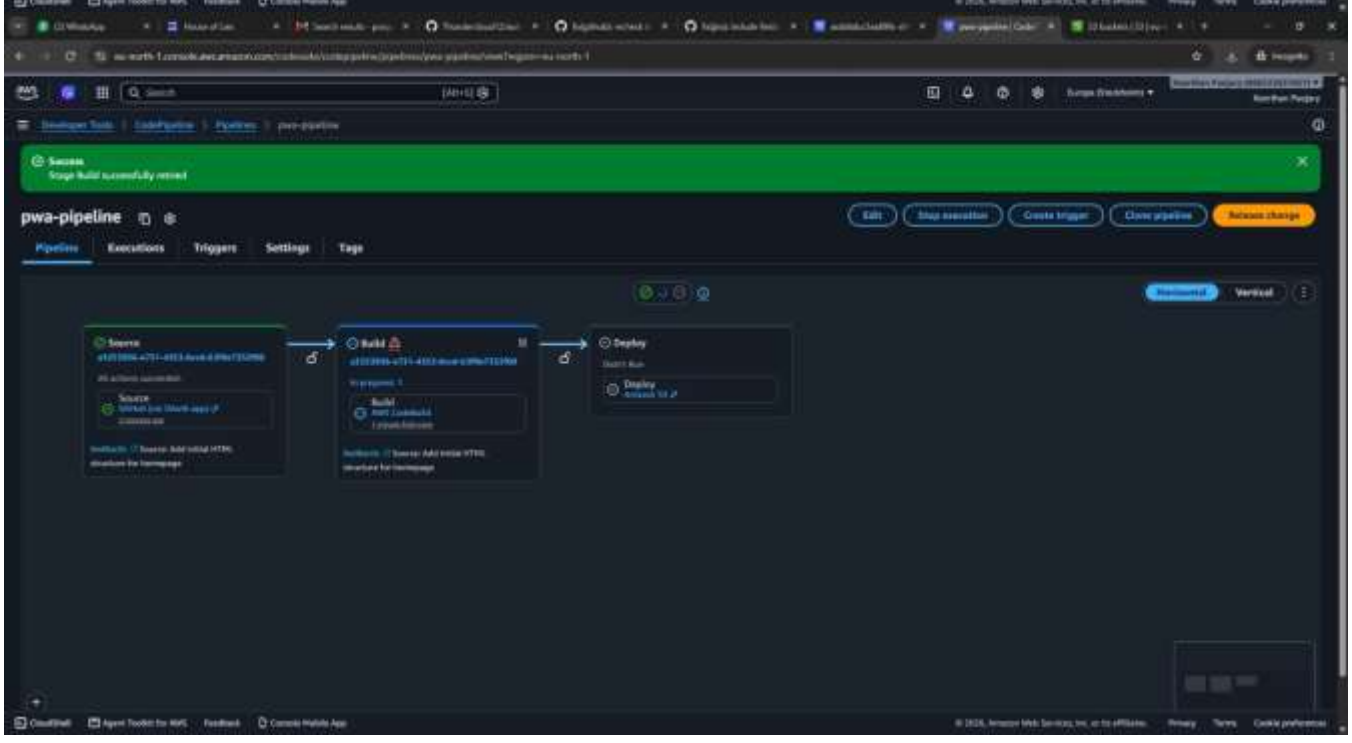
Working of the CI/CD Pipeline

The CI/CD pipeline follows these steps:

1. Code Development: The developer creates or modifies the website source code.
2. Code Repository: The source code is pushed to a GitHub repository.
3. Continuous Integration: GitHub Actions automatically triggers a workflow when new code is pushed.
4. Build and Testing: The workflow installs dependencies, builds the website if necessary, and runs automated tests.
5. Continuous Deployment: If the build and tests succeed, the workflow uploads the website files to an Amazon S3 bucket.
6. Website Hosting: Amazon S3 serves the deployed static website to users through its website endpoint. For production websites, Amazon CloudFront can provide HTTPS and content delivery.


Create build project
Project configuration
Project name
asddlab
Project type
Default project
Runner project
Additional configuration
Source
Source 1 - Primary
Source provider
GitHub
Credential
Use override credentials for this project only
Identity and Access Management (IAM)
Policy was successfully attached to role.
asddlab
Summary
Creation date
ARN
Last activity
Maximum session duration
1 hour
Permissions
Trust relationships
Tags
Last Accessed
Revoke sessions
Permissions policies (2)
Filter by Type
All types
Policy name
Type
Attached entities
AWS managed
Customer managed
Permissions boundary (not set)
Generate policy based on CloudTrail events


sotre-build-artifact-vb
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems
Access Points
Permissions overview
Block public access (bucket settings)
Block all public access
Individual Block Public Access settings for this bucket
Bucket policy
Public access is blocked because Block Public Access settings are turned on for this bucket


Success
pwa-pipeline
Edit
Stop execution
Create trigger
Clone pipeline
Release change
Pipeline
Executions
Triggers
Settings
Tags
Source
Build
Deploy



Welcome to My Website

- About
- Services
- Contact

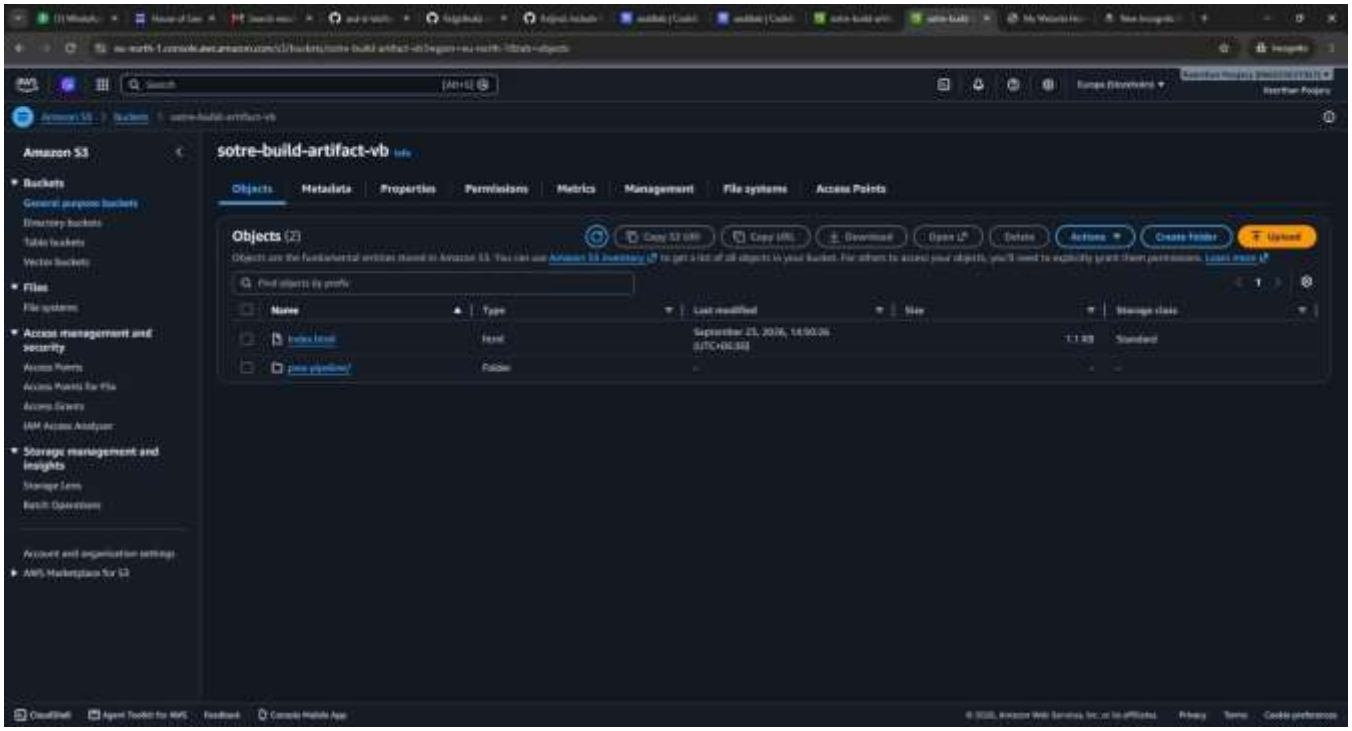
About Us

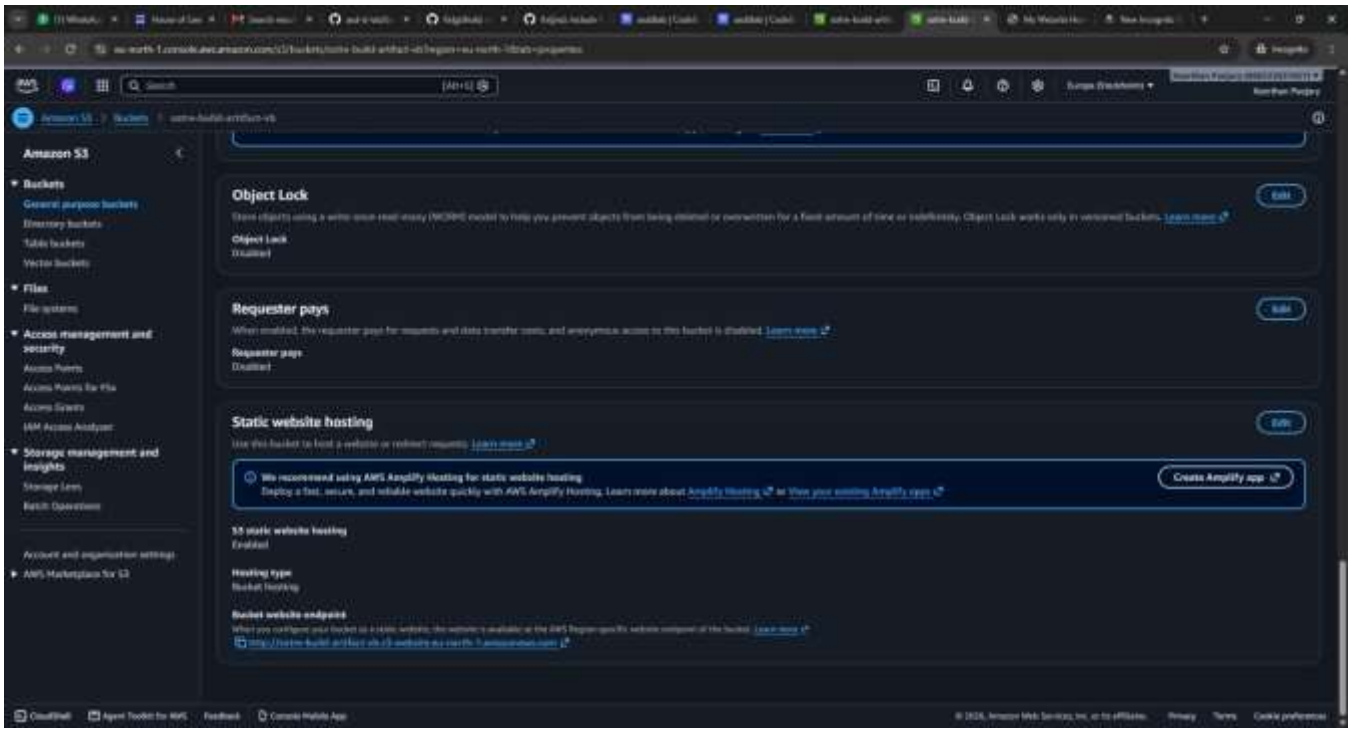
This is a basic paragraph describing the purpose of this webpage.

Our Services

- Web Design
- Development
- SEO Optimization

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Conclusion:

A CI/CD pipeline using GitHub Actions and AWS automates the process of building, testing, and deploying a website. By integrating version control with cloud hosting, developers can publish updates quickly and reliably. This approach improves development efficiency, reduces deployment errors, and demonstrates the practical use of cloud computing and DevOps principles